

# Computational Methods in Relativistic Hydrodynamics

## Applications to Astrophysical Mergers and Explosions

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National Conference on Computational Mathematics

*Special Session on Numerical Relativity*

July 3, 2026

- Relativistic hydrodynamics (RHD) plays a crucial role in modeling high-energy astrophysical phenomena.
- Examples include binary neutron star mergers<sup>1</sup>.
- The thermodynamic properties of a relativistic gas are often modeled using the Synge equation of state<sup>2</sup>.
- In extreme gravitational fields, we rely on Einstein's field equations to describe BNS mergers<sup>1</sup>.

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<sup>1</sup> L. Baiotti and L. Rezzolla. "Binary neutron star mergers: a review of Einstein's richest laboratory". In: *Reports on Progress in Physics* 80.9 (2017), p. 096901

<sup>2</sup> J. L. Synge and P. M. Morse. *The relativistic gas*. 1958

## Numerical Methods

### Theorem Conservation Laws

The governing equations of relativistic hydrodynamics can be written as a system of hyperbolic conservation laws:

$$\frac{\partial U}{\partial t} + \sum_{i=1}^3 \frac{\partial F^i(U)}{\partial x^i} = S(U)$$

where  $U$  is the vector of conserved variables.

### Definition Primitive Variables

The primitive variables (e.g., density  $\rho$ , velocity  $v$ , and pressure  $p$ ) must be recovered from the conserved variables  $U$  at each time step, typically requiring numerical root-finding methods.

## Simulation & Visualization

## Dynamics of RHD Shockwaves

- Relativistic shockwaves exhibit complex interface instability.
- High-order WENO schemes are employed to resolve the shocks without spurious oscillations.
- The animation on the right shows the evolution of the density field over 21 frames.

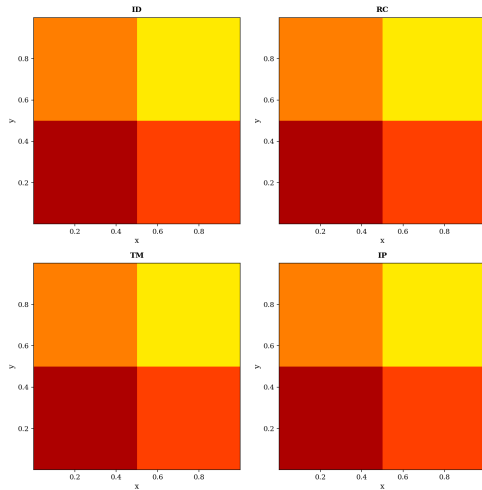


Figure: 2D Relativistic flow evolution.  **Mahindra**  
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- Robust numerical schemes are essential for simulating relativistic flows.
- The `mucomp` package provides support for standard theme styles, math blocks, footnoting/citations, and animations.